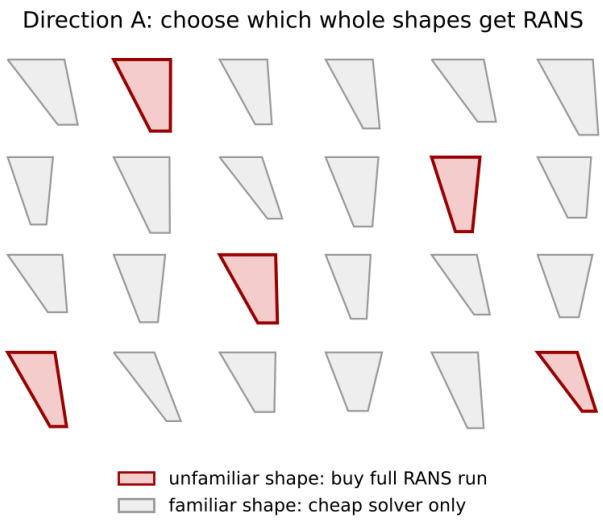
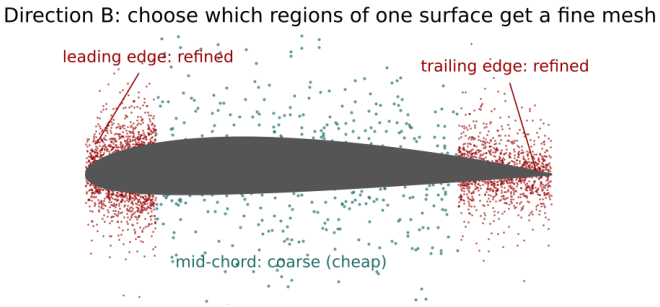
Case-Level vs Region-Level Multi-Fidelity: Which Direction?

Comparison of two directions for adaptive multi-fidelity mesh/graph surrogates, with the datasets usable for each. Prepared Nov 2026 for discussion with Prof. Nocera.

Bottom line. Direction A (choose which whole shapes get an expensive RANS run) is doable now on public data and is the safer core. **Direction B** (choose which regions of a surface, such as corners and edges, get a fine mesh) is more distinctive and closer to how engineers actually spend accuracy, but no public dataset supports it; it needs data we generate, starting with 2D airfoils. The two combine naturally: A decides *which shapes*, B decides *where on each shape*.





Schematics drawn for this comparison; not data.

Side by side

	Direction A: case-level	Direction B: region-level
Question	Which shapes need an accurate run?	Where on a shape is accuracy needed?
Unit of choice	One whole simulation (a wing, all 9 angles)	One region of one surface (leading edge, trailing edge, tips, kink)
How fidelity is varied	Cheap solver (VLM) vs RANS	Coarse vs fine mesh, region by region (local mesh refinement)
What is saved	Whole RANS runs (~127 CPU-h each on Shen & Alonso)	Mesh cells, so CPU time per run
Data available now	Yes: Shen & Alonso has both levels for 2,448 cases	No: needs generated runs with region-wise refinement
Compute needed	Laptop / small GPU (training only)	Hundreds to a few thousand 2D RANS runs, then training
CFD expertise needed	Low (use published data)	Medium-high (meshing, refinement, convergence checks)
Novelty	Med: adaptive multi-fidelity exists on grids; new on mesh/graph models	Med-High: adaptive mesh refinement is mature, but one surrogate guiding refinement across many shapes is less explored (needs a novelty check)
Risk	Gap between solvers may be too even to exploit	Regions interact (refining one changes the flow elsewhere); larger engineering effort
Fits professor's suggestion	Partly: picks unfamiliar shapes, not regions	Yes: exactly corners/edges vs common regions
Order	First: baseline + loop	Second: built on A's model and loop

Which is more doable?

Direction A is more doable now; Direction B is more distinctive but needs generated data.

	Direction A: case-level	Direction B: region-level	Multiphysics follow-up
Overall doability	High: start immediately	Medium: after a data-generation step	Medium-low: needs a matched cheap coupled model
First result	Days of work: B0 gap spread + B1 baseline on existing data	After the airfoil runs are generated and validated	After A works
Data to generate	None	~800-3,200 short 2D RANS runs	Cheap coupled runs only (fast)
Main blocker	Data format still to inspect	Meshing set-up + compute + validation	Reproducing the dataset's wing set-up
Novelty	Med	Med-High	High

Datasets usable for each direction

No public dataset gives region-wise fidelity for many shapes. Direction B therefore uses public data only for evidence and for single-geometry checks, and generated data for the main study.

Dataset	Direction A	Direction B	How it would be used
Shen & Alonso double-delta wings (2,448 cases, VLM + RANS)	Yes: main dataset	Partly: evidence only	A: the baseline and adaptive loop. B: surface uncertainty maps to show whether high uncertainty sits at edges, tips and the kink; no region-wise cost
Generated 2D airfoils with region-wise refinement (gmsh + OpenFOAM/SU2, AirfRANS-style setup)	Partly	Yes: main dataset for B	100-200 airfoils x 8-16 refinement patterns (each region coarse/fine), real cell counts and CPU time logged; loop chooses regions retrospectively
Setinek, Galletti, Brandstetter 2025 (611 airfoil pairs: wall functions vs resolved boundary layer)	Yes	Partly	A: second two-level check. B: a near-wall-only refinement is one specific "region" (the boundary layer); not released, ask authors
AIAA Drag Prediction Workshop grids (CRM, 6 grid sizes)	No	Partly	B: fidelity ladder on one aircraft; grids only, we would run CFD; one geometry
NASA Turbulence Modeling Resource (nested grid families)	No	Partly	B: validate our refinement and solver set-up against reference results
AirfRANS (1,000 airfoils, one fidelity)	Partly	Partly	Both: reuse its OpenFOAM set-up and meshes as the fine level for the generated study
BSCW multi-fidelity loads (3 levels, numbers only)	Partly	No	A: quick 3-level scoring check; no surface data, so not B
Transonic wing aerostructural (2,471, one fidelity)	Partly	No	A: multiphysics follow-up with a generated cheap level
FlowBench (10k+ shapes, 3 grid resolutions for 2D cavity sets)	Partly	Partly	Global resolution levels, not region-wise; grid-based
SuperWing, BlendedNet, AirfRANS, HiLiftAeroML	Partly	No	Single fidelity: backbone tests only

Yes = usable as the main data · Partly = supporting use only · No = not usable. Judgements are my own.

How each would run

Direction A: case-level loop	Direction B: region-level loop
1. Pool: 256 training wings; VLM visible for all. 2. Train a GNN with 5 output heads on the wings bought so far. 3. Score each unbought wing: spread of predicted lift and moment across heads, divided by its RANS cost. 4. Buy the top 4 (skip near-duplicates); reveal stored RANS; fine-tune. 5. Repeat to budget; compare with random, design order and best fixed mix at equal CPU-hours.	1. Split each airfoil into 3-4 regions (leading edge, trailing edge, upper, lower surface). 2. Precompute every refinement pattern per airfoil; log cells and CPU time. 3. GNN takes the coarse solution (+ refined regions so far) and predicts the fully refined surface field; heads give an uncertainty map. 4. Score each (airfoil, region): uncertainty in that region divided by the cost of refining it; reveal the stored run. 5. Compare with refine-all, refine-none, fixed rule (always leading edge) and random regions.

Related work to check (Direction B)

Region-wise accuracy is the territory of adaptive mesh refinement. Before committing, a novelty check should cover: classic adaptive refinement (Berger & Oliger 1984); adjoint-based output error estimation and adaptation (e.g. Venditti & Darmofal 2002); and recent learned refinement, including reinforcement-learning approaches (e.g. Yang et al. 2023; Foucart et al. 2023). The possible distinct angle: one surrogate, trained across many shapes, whose uncertainty decides where to refine on a new shape without solving an error estimate per shape. These references are named for checking, not yet reviewed in detail.

Recommendation

1. Start with A on Shen & Alonso. Baseline, uncertainty heads and the case-level loop. Fully doable now on published data.

2. Add the cheap bridge to B on the same data: surface uncertainty maps. If uncertainty concentrates at leading edges, tips and the kink, that is direct evidence for the professor's idea, at no extra compute.

3. Then build B on generated 2D airfoils (region-wise refinement patterns, AirfRANS-style set-up, validated against NASA verification cases). This becomes the main novel study.

4. Combined claim, if both work: a surrogate that decides which shapes need accurate simulation (A) and where on each shape (B).

Questions for the professor

1. Should region-level (B) be the main contribution, with case-level (A) as the baseline it is compared against?

2. Is generating 2D airfoil runs with region-wise refinement acceptable, and can we get USC compute for a few thousand short runs?

3. Is there a colleague with mesh-adaptation or CFD experience who could advise on the refinement set-up?

Datasets: size and doability per direction

Easy = use as is · Moderate = some processing or generation · Hard = large effort or compute · No = not usable for that direction. Sizes as stated by each source; "not stated" where none was given.

Dataset	Cases	Size	A: case-level	B: region-level	Multiphysics
Shen & Alonso double-delta wings	2,448 (272 wings x 9 angles)	Not stated; volume files dominate, surface-only much smaller	Easy	Moderate: maps only	No

Dataset	Cases	Size	A: case-level	B: region-level	Multiphysics
Generated 2D airfoils, region-wise refinement	~100-200 airfoils x 8-16 patterns (800-3,200 runs)	We create it (est.)	Moderate	Moderate	No
Setinek, Galletti, Brandstetter 2025	611 pairs	7.8 GB + 18 GB	Moderate: ask authors	Moderate: near-wall only	No
AIAA DPW-7 grids (CRM)	1 geometry, 6 grid sizes	0.4-13 GB per grid	No	Hard: we run the CFD	No
NASA Turbulence Modeling Resource	A few verification cases	Small	No	Easy: validation only	No
AirfRANS	1,000	Not stated here	Moderate: add a cheap level	Moderate: reuse set-up	No
BSCW multi-fidelity loads	732 wing + 2,250 eVTOL	Under 20 MB	Easy: 3-level check	No	No
Transonic wing aerostructural	2,471 + 3 OOD	Not stated	Moderate	No	Moderate: generate cheap level
FlowBench	10,000+	1.72 TB	Hard	Hard: global levels only	Moderate: flow + heat, grids
SuperWing	28,856 flow fields	~3-23 GB processed	No	No	No
BlendedNet	8,830	~50 GB	No	No	No
HiLiftAeroML	1,800	~63 TB	No	No	No

Short version: for A, Shen & Alonso is the one dataset to download now. For B, the main data must be generated (2D airfoils), with NASA verification cases to validate it. For multiphysics, the aerostructural dataset plus a generated cheap level is the most realistic route.

Sources

Shen & Alonso 2025, arXiv 2512.20941 (dataset) · Setinek, Galletti, Brandstetter 2025, arXiv 2511.01830 · AIAA DPW-7 grids, dpw.larc.nasa.gov/DPW7 · NASA Turbulence Modeling Resource · Bonnet et al. 2022, AirfRANS, arXiv 2212.07564 · Vaiuso, Immordino, Righi, Da Ronch 2025, zenodo.org/records/17104476 · Koratikere, Dikshit, Leifsson 2026, arXiv 2610.36442 · FlowBench (Hugging Face BGLab/FlowBench) · Yang et al. 2025, SuperWing · Sung et al. 2025, BlendedNet · Ashton et al. 2026, HiLiftAeroML · Berger & Oliger 1984; Venditti & Darmofal 2002; Yang et al. 2023; Foucart et al. 2023 (to review).